

AAKASH GHOSH

Generative AI & Agentic AI Engineer

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PROFESSIONAL SUMMARY

Engineer with **6+ years** of experience delivering validated enterprise QMS systems in pharmaceutical and life sciences, now focused on **Generative AI** and **Agentic AI** engineering. Has built hands-on **RAG** and **multi-agent** solutions that automate real support and operations workflows, and is **Claude Foundations certified (Anthropic)**. Combines **LLM**, **RAG**, and **agent orchestration** skills with deep **21 CFR Part 11** knowledge, bringing a compliance-first, audit-ready approach to AI. Looking to build production-grade GenAI and agentic AI solutions for enterprise and regulated environments.

CORE SKILLS

Generative AI & LLMs	Generative AI, Large Language Models (LLMs), Claude (Anthropic), Google Gemini, Prompt Engineering, Context Grounding
RAG & Retrieval	Retrieval-Augmented Generation (RAG), Document Ingestion & Parsing, Chunking Strategies, Embeddings, Sentence-Transformers, ChromaDB, Vector Databases, Semantic Search
Agentic AI	AI Agents, Multi-Agent Systems, Agent Orchestration, LangGraph, LangChain, Human-in-the-Loop Design
Integration & APIs	Python, AWS Bedrock, Oracle SQL
Enterprise QMS Platforms	TrackWise, ValGenesis VLMS

CERTIFICATIONS

- Claude Certified Architect – Foundations, Anthropic

PROFESSIONAL EXPERIENCE

Cognizant Technology Solutions, Kolkata, India

Nov 2019 – Present

I. Generative AI & Agentic AI

Apr 2026 – Present

- Delivered **2 GenAI and agentic AI systems** (RAG chatbot, agentic DS traceability pipeline) that cut support-ticket volume by **47%** and manual compliance-documentation effort - reducing operational overhead and risk across the QMS program.

II. TrackWise Developer

Mar 2020 – Present

Project: Enterprise QMS Modernization Program | Client: Philips | Domain: Healthcare

- Configuration & development:** Designed and customized **12+ TrackWise workflows and business rules** for the Complaint Handling module, aligned to client SOPs and compliant with **21 CFR Part 11**.
- Requirements:** Led **20+ requirement workshops** with global business users and quality teams, with **zero requirement-change rework** after sign-off.
- Validation (ValGenesis):** Managed the validation lifecycle for **4+ years**; authored and reviewed **20+ IQ / PIQ / MP documents** and led **IQ / PIQ execution** for multiple major releases with a **100% first-pass audit record**.
- Testing, deployment & support:** Coordinated QA/UAT for 6+ feature deployments (**40+ defects** resolved before go-live); managed controlled Dev, QA, and Production releases with **zero unplanned downtime**; provided post-production support and upgrade work.

PROJECTS

Project 1: DS Traceability Copilot

Tech Stack: Python, LangGraph, Google Gemini, ChromaDB, Sentence-Transformers, Slack Bolt, FastAPI, Google Drive API

- Built a **multi-agent LangGraph pipeline** linking Slack user stories to **Design Specification (DS) IDs** in a **GxP-regulated TrackWise complaint-handling document**: a **matcher agent** (semantic search via ChromaDB + Sentence-Transformers), a **Gemini drafting agent** producing house-style DS entries when no match exists, and a **self-correction reviewer agent** that critiques and revises drafts before human review.
- Business Impact:** Automated DS traceability mapping and drafting for a GxP-regulated document, improving compliance accuracy and audit-readiness and removing manual cross-referencing effort

Project 2: RAG-Based Support Chatbot for Business Users

Tech Stack: Python, RAG, Embeddings, Vector Database, LLM, Ollama, LangChain, Streamlit

- Built a **RAG chatbot** over the project's full documentation set (**SOPs, work instructions, Design Specifications, DF sheets**), with a **document ingestion, chunking, embedding, and retrieval pipeline** grounding responses in approved documents rather than the model's general knowledge.
- Business Impact:** Reduced process-related support tickets by **47%**, cutting resolution time and freeing the support team for higher-value work.

EDUCATION

M.Tech, Software Systems, BITS Pilani (WILP)

Graduated 2024

B.Tech, Electronics and Communication, College of Engineering & Management (CEM), Kolaghat

Graduated 2019